

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-13	Team No- 06	Project Title- The Chronic Kidney Disease (CKD) Patient Management System
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1. Abstract

The **Chronic Kidney Disease (CKD) Patient Management System** is a database-driven healthcare application designed to efficiently manage and organize patient medical information. Managing patient records, laboratory test results, doctor details, appointments, and CKD prediction results manually can lead to data duplication, inconsistency, and difficulty in retrieving important information. The proposed system provides a centralized platform for storing and managing healthcare data using relational database concepts such as ER modeling, normalization, constraints, indexing, joins, views, procedures, triggers, and transactions. It maintains structured information about patients, doctors, medical tests, appointments, and prediction records while ensuring data integrity and consistency. The system can also integrate MongoDB for semi-structured data and provide backend services using FastAPI and Pydantic. Security mechanisms such as JWT, OAuth2, and role-based access control (RBAC) can be used to protect sensitive information. The proposed solution aims to provide a secure, reliable, and efficient database system that simplifies healthcare data management and demonstrates the practical application of modern DBMS and database engineering concepts.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520080060	J.Sohan	
2520090137	J.Sri charan	
2520080036	Adithya vishnubatala	

3. Frameworks / Platforms / Software Used

PostgreSQL, MongoDB, MongoDB BSON, FastAPI, Pydantic, SQLAlchemy, Motor, PyMongo, Flask, Django, Django REST Framework (DRF), Node.js, Express.js, NestJS, Spring Boot, Spring Data JPA, .NET Core, ASP.NET Core, Entity Framework Core (EF Core), Redis, Pinecone, pgvector, Docker, Docker Compose, Kubernetes, GitHub, GitHub Actions, Prometheus, Grafana, OpenTelemetry, Postman, SQLAlchemy Testing, pytest, Coverage Tools, OpenAPI, Swagger.

Guide Name and Signature (remarks if any) _____